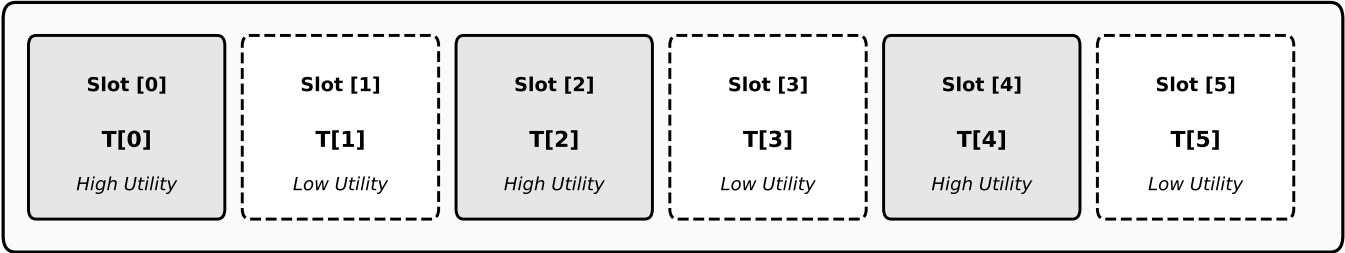


DYNAMIC KEY-VALUE (KV) CACHE IN-SITU STATE TRANSFORMATION

STATE A: FRAGMENTED KV-CACHE BUFFER (PRIOR TO COMPACTION)



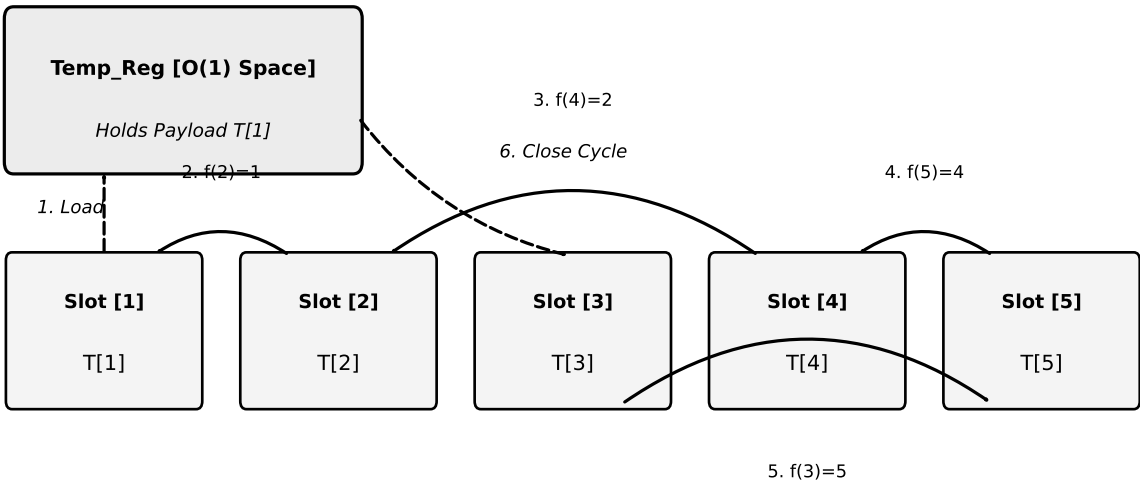
ASSOCIATIVE TARGET MAP  $f(x)$  & IN-PLACE DISJOINT CYCLE TRAVERSAL

**Idempotent Invariant:**  $f(f(x)) = f(x)$  [Attractor Basin Locking]

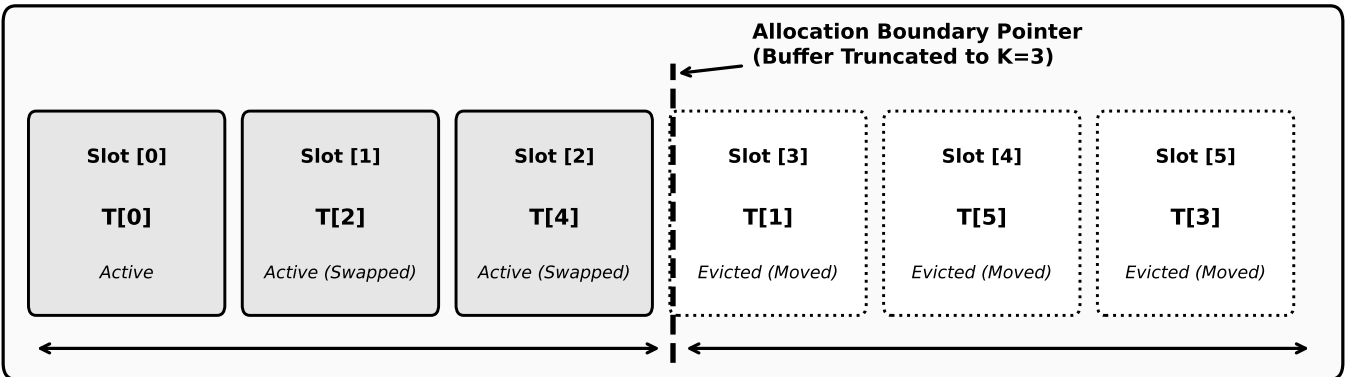
\* Fixed Point:  $f(0) = 0$  [Retained at Origin - Zero Data Movement]

\* Permutation Mapping:  $f(1)=3, f(2)=1, f(3)=5, f(4)=2, f(5)=4$

\* Disjoint Orbit: ( 1 --> 3 --> 5 --> 4 --> 2 --> 1 )



STATE B: IN-SITU COMPACTED CONTIGUOUS BUFFER (POST-TRUNCATION)



Contiguous Active Working Context ( $K=3$ )  
[Direct Hardware Coalescing / FlashAttention]

Released / Truncated Memory Region  
[Overwritten on Subsequent Inferences]